Ngao seasonal outlook: El Niño & Indian Ocean model for Kenya's short rains

Team Daraja · GreenTech Hackathon 2026 · Ngao / MvuaSmart

Data (all real, open, downloaded 14–15 Oct 2026)

• **Rainfall:** CHIRPS v2.0 monthly (UCSB CHC), Jan 1981–Aug 2026, averaged over **1,452 wards** (geoBoundaries KEN ADM3). **GPCC** Full Data v2020 (1951–2019) extends training back to 1951. Over 1981–2015 it agrees with CHIRPS: Kenya-mean correlation 0.989, 86.7% tercile label agreement.
• **Ocean/atmosphere:** NOAA Niño 3.4, ONI and DMI (IOD) indices; NOAA ERSSTv5 SST; NCEP/NCAR Reanalysis 1 850 hPa winds and sea-level pressure (NOAA PSL).
• **Sources:** data.chc.ucsb.edu/products/CHIRPS-2.0 · psl.noaa.gov · cpc.ncep.noaa.gov · geoboundaries.org. No synthetic data.

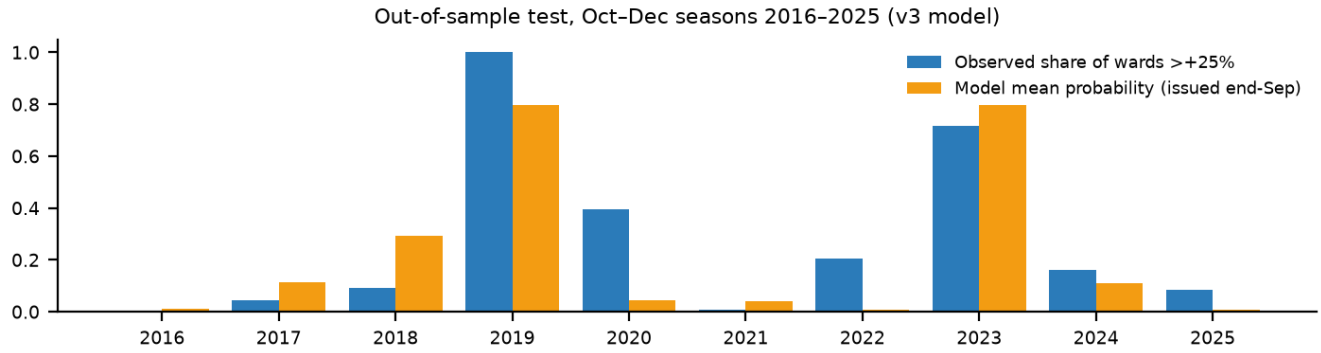
Model

Regularised logistic regression on **6 literature indices**: western and eastern Indian Ocean SST, the Indian Ocean Dipole, Niño 3.4, central equatorial Indian Ocean 850 hPa winds, and the east–west Indian Ocean pressure gradient (Saji 1999; Black 2003; Behera 2005; Hastenrath 2011; Nicholson 2017), plus region dummies. **Issued end of September** for the Oct–Dec short rains, using Jun–Sep observations only (no leakage). Trained 1951–2015, tested on unseen seasons 2016–2025. Hyperparameters and alert thresholds were chosen by training-only, year-blocked cross-validation.

Scores on the test set, 2016–2025 (ward-seasons)

Target / model	ROC-AUC (90% CI)	F1	Brier
Wet season >+25% vs normal: v3 (this model)	**0.775 (0.42–0.90)**	**0.662**	**0.122**
v2: logistic on 6 Niño/IOD indices	0.726	0.634	0.163
ONI rule (El Niño Jul–Sep ≥ 0.5 → wet)	0.720	0.388	–
KMD tercile "above normal": v3 (this model)	**0.736 (0.52–0.87)**	**0.642**	**0.200**
v2: logistic on 6 indices	0.700	0.642	0.212
ONI rule	0.746*	0.323	–

*ONI rule AUC scored on raw ONI values. Higher AUC/F1 is better; lower Brier is better (0 = perfect probabilities).



Honest limits

• All wards in a season move together, so the test is effectively **~10 seasons**. The AUC 90% CI is roughly **0.4–0.9**, and the gains over v2 are directional, not statistically proven (most of the gain comes from removing the 2018 false alarm).
• It caught 2019 and 2023, but **missed the wet signals of 2020, 2022, 2024 and 2025**; June–September ocean conditions don't explain those seasons.
• Dynamical forecasts (C3S, NMME) needed credentials and weren't used. This is a decision-support layer, **not a replacement for the official KMD/ICPAC outlook**.

Next steps

• Add Copernicus C3S seasonal hindcasts and ERA5 (needs a CDS API key) to blend dynamical forecasts with the statistical model.
• Run a live end-of-September outlook per ward in **Oracle Autonomous Database + APEX**, pushed with Ngao's SMS/USSD alerts in Swahili/English.